THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Sequences and Progressions** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 50	$d = 3$, $a = 5$; $t_{16} = a + 15d$.
2	**(a)** 37	$t_k = S_k - S_{k-1} = p(2k-1) + q$.
3	**(c)** 816	AP with $a = 6$, $d = 6$.
4	**(a)** 7	$\frac{n}{2}[2a + (n-1)d] = 510$ gives d.
5	**(b)** 14	Take $m - d,\ m,\ m + d$; the sum gives the middle number.
6	**(a)** 12100	Yearly values form a GP with ratio $1 + \frac{r}{100}$.
7	**(b)** 2	$r^7 = 128$.
8	**(d)** 440	Add the first and last term, multiply by $\frac{n}{2}$.
9	**(a)** Both A and R are true and R is the correct explanation of A.	$5 - 2 = 8 - 5 = 11 - 8 = 3$; a constant common difference is the definition of an AP.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	The ratio of consecutive terms of the first is 2 and the difference of consecutive terms of the second is 2; both are true, but they are unrelated statements.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\frac{n}{2}[2a + (n-1)d] = 255$; $d = 2$ (1); $t_{15} = 31$ (1).	1+1
12	$a + (n-1)d > 200$; $(n-1) > 32.50$ (1); $n = 34$ (1).	1+1
13	$t_1 = S_1 = 7$ (1); $t_{10} = S_{10} - S_9 = 61$ (1).	1+1
14	$\frac{n}{2}[2a + (n-1)d] = 336$; $d = -4$ (1); $t_{12} = 6$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$2a + 10d = 40$ and $a + 4d = 17$ (1); Solving: $a = 5$, $d = 3$ (1); $S_{20} = 670$ (1).	1+1+1
16	Terms: $12, 17, 22, \ldots, 197$ (1); Number of terms $= 38$ (1); Sum $= \frac{n}{2}(a + l) = 3971$ (1).	1+1+1
17	Let them be $\frac{m}{r},\ m,\ mr$; Product $= m^3 = 216$, so $m = 6$ (1); $\frac{6}{r} + 6r = 15$ gives $r = 2$ (1); Numbers 3, 6, 12 (1).	1+1+1
	OR: $2a + 10d = 64$ and $a + 4d = 28$ (1); Solving: $a = 12$, $d = 4$ (1); $S_{14} = 532$ (1).	1+1+1
18	Terms: $24, 31, 38, \ldots, 297$ (1); Number of terms $= 40$ (1); Sum $= \frac{n}{2}(a + l) = 6420$ (1).	1+1+1
	OR: Let them be $\frac{m}{r},\ m,\ mr$; Product $= m^3 = 1728$, so $m = 12$ (1); $\frac{12}{r} + 12r = 40$ gives $r = 3$ (1); Numbers 4, 12, 36 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $a = 100,\ d = 50$ (1). (ii) $t_{12} = 100 + 11 \times 50 = 650$ (1). (iii) $S_{12} = \frac{12}{2}(100 + 650) = 6 \times 750 = 4500$ (1). (iv) $100 + (n-1)50 = 1000 \Rightarrow n - 1 = 18 \Rightarrow n = 19$ (2).	1+1+1+2
20	Penalties form an AP (1); $t_{24} = 100 + 23 \times 25 = 675$ (1); $S_n = \frac{n}{2}(a + l)$ (1); $S_{24} = \frac{24}{2}(100 + 675)$ (1); Total ₹9300 (1).	1+1+1+1+1
	OR: Ratio $\frac{3}{4}$ is constant, so a GP (1); 1st rebound $= 48$ m (1); 3rd rebound $= h \times (3/4)^3$ (1); $= 27$ m (2).	1+1+1+2

Mistakes to avoid:

- Writing $t_n = a + nd$ instead of $a + (n-1)d$, which makes the first term wrong.
- Forgetting the factor $\frac{n}{2}$ in S_n, or using $n(n+1)$ for the sum of the first n natural numbers (it is $\frac{n(n+1)}{2}$).
- Using $t_n = ar^n$ for a GP instead of ar^{n-1}.
- Calling a sequence an AP after checking only the first two terms; the difference must be the same for all consecutive terms.
- Taking S_n of an AP to be the nth term, or finding the nth term as S_n instead of $S_n - S_{n-1}$.
- Sign errors with a negative common difference or ratio, e.g. $t_5 = 7 + 4 \times (-2) = -1$, not $7 + 4 \times 2$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Sequences and Progressions** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 2812	AP with $a = 4$, $d = 4$.
2	**(c)** -3	$\frac{n}{2}[2a + (n-1)d] = 147$ gives d.
3	**(a)** 17	Take $m - d,\ m,\ m + d$; the sum gives the middle number.
4	**(a)** 28800	Yearly values form a GP with ratio $1 + \frac{r}{100}$.
5	**(d)** 3	$r^5 = 243$.
6	**(b)** 319	Add the first and last term, multiply by $\frac{n}{2}$.
7	**(d)** 97	$d = 9$, $a = 7$; $t_{11} = a + 10d$.
8	**(d)** 57	$t_k = S_k - S_{k-1} = p(2k-1) + q$.
9	**(c)** A is true but R is false.	$\frac{10 \times 11}{2} = 55$, so A is true; R is false because the sum is $\frac{n(n+1)}{2}$.
10	**(d)** A is false but R is true.	$\frac{6}{3} = 2$, $\frac{12}{6} = 2$, $\frac{18}{12} = 1.5$; the ratio is not constant, so A is false and R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$a + (n-1)d > 150$; $(n-1) > 19.86$ (1); $n = 21$ (1).	1+1
12	$t_1 = S_1 = 7$ (1); $t_{12} = S_{12} - S_{11} = 51$ (1).	1+1
13	$\frac{n}{2}[2a + (n-1)d] = 1218$; $d = 5$ (1); $t_{21} = 108$ (1).	1+1
14	$a + (n-1)d > 301$; $(n-1) > 33.22$ (1); $n = 35$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$2a + 10d = 62$ and $a + 6d = 38$ (1); Solving: $a = -4$, $d = 7$ (1); $S_{12} = 414$ (1).	1+1+1
16	Terms: $17, 21, 25, \ldots, 149$ (1); Number of terms $= 34$ (1); Sum $= \frac{n}{2}(a + l) = 2822$ (1).	1+1+1
17	Let them be $\frac{m}{r},\ m,\ mr$; Product $= m^3 = 8000$, so $m = 20$ (1); $\frac{20}{r} + 20r = 85$ gives $r = 4$ (1); Numbers 5, 20, 80 (1).	1+1+1
	OR: $2a + 14d = 32$ and $a + 8d = 14$ (1); Solving: $a = 30$, $d = -2$ (1); $S_{16} = 240$ (1).	1+1+1
18	Terms: $31, 40, 49, \ldots, 247$ (1); Number of terms $= 25$ (1); Sum $= \frac{n}{2}(a + l) = 3475$ (1).	1+1+1
	OR: Let them be $\frac{m}{r},\ m,\ mr$; Product $= m^3 = 3375$, so $m = 15$ (1); $\frac{15}{r} + 15r = 50$ gives $r = 3$ (1); Numbers 5, 15, 45 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $d = 2$ (1). (ii) $t_{15} = 20 + 14 \times 2 = 48$ (1). (iii) $S_{15} = \frac{15}{2}(20 + 48) = 510$ (1). (iv) $S_{10} = \frac{10}{2}(40 + 18) = 290$ (1); last 5 rows $= 510 - 290 = 220$ seats (1).	1+1+1+2
20	Penalties form an AP (1); $t_{28} = 150 + 27 \times 30 = 960$ (1); $S_n = \frac{n}{2}(a + l)$ (1); $S_{28} = \frac{28}{2}(150 + 960)$ (1); Total ₹15540 (1).	1+1+1+1+1
	OR: Ratio $\frac{3}{4}$ is constant, so a GP (1); 1st rebound $= 192$ m (1); 3rd rebound $= h \times (3/4)^3$ (1); $= 108$ m (2).	1+1+1+2

Mistakes to avoid:

- Writing $t_n = a + nd$ instead of $a + (n-1)d$, which makes the first term wrong.
- Forgetting the factor $\frac{n}{2}$ in S_n, or using $n(n+1)$ for the sum of the first n natural numbers (it is $\frac{n(n+1)}{2}$).
- Using $t_n = ar^n$ for a GP instead of ar^{n-1}.
- Calling a sequence an AP after checking only the first two terms; the difference must be the same for all consecutive terms.
- Taking S_n of an AP to be the nth term, or finding the nth term as S_n instead of $S_n - S_{n-1}$.
- Sign errors with a negative common difference or ratio, e.g. $t_5 = 7 + 4 \times (-2) = -1$, not $7 + 4 \times 2$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Sequences and Progressions** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 22	Take $m-d,\ m,\ m+d$; the sum gives the middle number.
2	**(c)** 6655	Yearly values form a GP with ratio $1+\frac{r}{100}$.
3	**(a)** 2	$r^8 = 256$.
4	**(a)** 560	Add the first and last term, multiply by $\frac{n}{2}$.
5	**(a)** 57	$d=6$, $a=-15$; $t_{13}=a+12d$.
6	**(a)** 35	$t_k = S_k - S_{k-1} = p(2k-1)+q$.
7	**(d)** 3588	AP with $a=13$, $d=13$.
8	**(d)** 4	$\frac{n}{2}[2a+(n-1)d]=640$ gives d.
9	**(a)** Both A and R are true and R is the correct explanation of A.	$3+9\times4=39$, using R.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	$7+4\times(-2)=-1$, so A is true; R is true but only says a negative difference is allowed, it does not give the value.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$t_1 = S_1 = 6$ (1); $t_8 = S_8 - S_7 = 76$ (1).	1+1
12	$\frac{n}{2}[2a+(n-1)d]=297$; $d=3$ (1); $t_{18}=42$ (1).	1+1
13	$a+(n-1)d>111$; $(n-1)>24.25$ (1); $n=26$ (1).	1+1
14	$t_1 = S_1 = 11$ (1); $t_{15} = S_{15} - S_{14} = 123$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$2a+8d=64$ and $a+9d=62$ (1); Solving: $a=8$, $d=6$ (1); $S_{13}=572$ (1).	1+1+1
16	Terms: $11, 17, 23, \ldots, 179$ (1); Number of terms $=29$ (1); Sum $=\frac{n}{2}(a+l)=2755$ (1).	1+1+1
17	Let them be $\frac{m}{r},\ m,\ mr$; Product $=m^3=13824$, so $m=24$ (1); $\frac{24}{r}+24r=60$ gives $r=2$ (1); Numbers 12, 24, 48 (1).	1+1+1
	OR: $2a+16d=84$ and $a+3d=17$ (1); Solving: $a=2$, $d=5$ (1); $S_{25}=1550$ (1).	1+1+1
18	Terms: $43, 51, 59, \ldots, 259$ (1); Number of terms $=28$ (1); Sum $=\frac{n}{2}(a+l)=4228$ (1).	1+1+1
	OR: Let them be $\frac{m}{r},\ m,\ mr$; Product $=m^3=27000$, so $m=30$ (1); $\frac{30}{r}+30r=156$ gives $r=5$ (1); Numbers 6, 30, 150 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $r=3$ (1). (ii) $t_5=3^4=81$ (1). (iii) $t_n=3^{n-1}$ (1). (iv) $3^5=243<500$ and $3^6=729>500$, so $n-1=6$, i.e. stage 7 (2).	1+1+1+2
20	Penalties form an AP (1); $t_{22}=250+21\times40=1090$ (1); $S_n=\frac{n}{2}(a+l)$ (1); $S_{22}=\frac{22}{2}(250+1090)$ (1); Total ₹14740 (1).	1+1+1+1+1
	OR: Ratio $\frac{2}{3}$ is constant, so a GP (1); 1st rebound $=108$ m (1); 3rd rebound $=h\times(2/3)^3$ (1); $=48$ m (2).	1+1+1+2

Mistakes to avoid:

- Writing $t_n = a+nd$ instead of $a+(n-1)d$, which makes the first term wrong.
- Forgetting the factor $\frac{n}{2}$ in S_n, or using $n(n+1)$ for the sum of the first n natural numbers (it is $\frac{n(n+1)}{2}$).
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